Exercise 1: Convergence and L^p spaces

Problem statement | 7 points | Suggested time: 25 minutes

Consider the measure space

$$\big((0,1), \mathcal{B}((0,1)), \lambda\big),$$

where λ denotes Lebesgue measure. For each integer $n \geq 1$, define

$$f_n(x) = \sqrt{n}\, \mathbf{1}_{(0,1/n)}(x), \qquad x \in (0,1).$$

Here $\mathbf{1}_A$ is the indicator function of the set A.

Answer all parts and justify your conclusions.

(a) **[2 points]** Determine the pointwise limit of (f_n) and compute

$$\int_0^1 f_n(x)\, dx.$$

Does the limit of the integrals equal the integral of the pointwise limit?

(b) **[3 points]** Compute $\|f_n\|_{L^1(0,1)}$ and $\|f_n\|_{L^2(0,1)}$. Determine whether f_n converges to the zero function in $L^1(0,1)$ and in $L^2(0,1)$.

(c) **[2 points]** Prove that, for every $n \geq 1$ and every $x \in (0,1)$,

$$|f_n(x)| \leq \frac{1}{\sqrt{x}}.$$

Use this bound to justify interchanging the limit and the integral by applying an appropriate convergence theorem. Explicitly verify its hypotheses.

Exercise 1: Detailed solution

Convergence and L^p spaces

(a) Pointwise limit and integrals. Fix $x \in (0,1)$. Choose an integer $N > 1/x$. For every $n \geq N$, we have $1/n < x$, so $x \notin (0, 1/n)$ and $f_n(x) = 0$. Thus

$$f_n(x) \longrightarrow f(x) = 0 \qquad \text{for every } x \in (0,1).$$

The function f_n is constant with value $\sqrt{n}$ on an interval of length $1/n$ and is zero elsewhere. Hence

$$\int_0^1 f_n(x)\,dx = \sqrt{n}\,\lambda((0,1/n)) = \sqrt{n}\,\frac{1}{n} = \frac{1}{\sqrt{n}} \longrightarrow 0.$$

Since $f = 0$, its integral is also zero. Therefore

$$\boxed{\lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \int_0^1 \lim_{n\to\infty} f_n(x)\,dx = 0.}$$

(b) Convergence in L^1 and L^2. For $1 \leq p < \infty$, the norm is $\|h\|_{L^p} = (\int_0^1 |h(x)|^p\,dx)^{1/p}$. As $f_n \geq 0$, the calculation in part (a) gives

$$\|f_n\|_{L^1} = \int_0^1 |f_n(x)|\,dx = \frac{1}{\sqrt{n}}.$$

For the L^2 norm, we must first square the function:

$$\|f_n\|_{L^2}^2 = \int_0^1 |f_n(x)|^2\,dx = \int_0^{1/n} n\,dx = 1, \qquad \|f_n\|_{L^2} = 1.$$

Both norms are finite for every n, so $f_n \in L^1(0,1) \cap L^2(0,1)$. Convergence to zero in L^p means $\|f_n - 0\|_{L^p} \to 0$. Consequently,

$$\boxed{f_n \to 0 \text{ in } L^1(0,1), \qquad f_n \not\to 0 \text{ in } L^2(0,1).}$$

This illustrates that pointwise convergence does not, by itself, imply convergence in L^2.

(c) Application of the dominated convergence theorem. If $0 < x < 1/n$, then $n < 1/x$, so $|f_n(x)| = \sqrt{n} \leq x^{-1/2}$. If $x \geq 1/n$, then $f_n(x) = 0$, and the same inequality holds. Thus the bound is valid for all $x \in (0,1)$ and all n.

Set $g(x) = x^{-1/2}$. This function is independent of n, measurable, and integrable because

$$\int_0^1 g(x)\,dx = \lim_{\varepsilon \downarrow 0} \big[2\sqrt{x}\big]_\varepsilon^1 = 2 < \infty.$$

Each f_n is measurable, $f_n \to 0$ everywhere, and $|f_n| \leq g \in L^1(0,1)$. The dominated convergence theorem therefore yields

$$\lim_{n\to\infty} \int_0^1 f_n\,d\lambda = \int_0^1 0\,d\lambda = 0.$$

There is no contradiction with part (b): applying the theorem to $|f_n|^2$ would require an integrable bound for the squared functions. The bound $g^2(x) = 1/x$ is not integrable on $(0,1)$.

Exercise 2: Joint and conditional distributions

Problem statement | 7 points | Suggested time: 25 minutes

Let X and Y be random variables with joint probability density

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Answer all parts and justify your conclusions.

(a) **[2 points]** Determine the marginal densities of X and Y, specifying their supports. Are X and Y independent? Justify your answer.

(b) **[3 points]** Determine the conditional density of Y given $X = x$, for $0 < x < 1$, and compute

$$\mathbb{E}[Y \mid X = x].$$

Then express the conditional expectation $\mathbb{E}[Y \mid X]$ as a function of the random variable X.

(c) **[2 points]** Compute $\mathbb{E}[Y]$ using the tower property of conditional expectation. Check your answer by integrating against the marginal density of Y.

Exercise 2: Detailed solution

Joint and conditional distributions

(a) Marginal densities and independence. The support is the triangle below the line $y = x$ inside the unit square. For a fixed $x \in (0,1)$, the admissible values of y lie in $(0, x)$. Therefore

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dy = \int_0^x 2\,dy = 2x, \qquad 0 < x < 1.$$

For a fixed $y \in (0,1)$, the admissible values of x lie in $(y, 1)$. Hence

$$f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dx = \int_y^1 2\,dx = 2(1-y), \qquad 0 < y < 1.$$

Both densities are zero outside $(0,1)$. As a check, $\int_0^1 2x\,dx = 1$ and $\int_0^1 2(1-y)\,dy = 1$.

Independence would require $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for Lebesgue-almost every (x,y). On the entire triangle $0 < x < y < 1$, however,

$$f_{X,Y}(x,y) = 0, \qquad f_X(x)f_Y(y) = 4x(1-y) > 0.$$

This triangle has positive area, so the equality fails on a set of positive measure. Therefore, X **and** Y **are not independent**.

(b) Conditional density and conditional expectation. For $0 < x < 1$, the marginal density $f_X(x) = 2x$ is positive. A version of the conditional density is

$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \begin{cases} 1/x, & 0 < y < x, \\ 0, & \text{otherwise.} \end{cases}$$

It integrates to one, since $\int_0^x (1/x)\,dy = 1$. In particular, the conditional distribution is uniform on $(0, x)$. Its mean is

$$\mathbb{E}[Y \mid X = x] = \int_{\mathbb{R}} y f_{Y|X=x}(y)\,dy = \frac{1}{x}\int_0^x y\,dy = \frac{1}{x}\frac{x^2}{2} = \frac{x}{2}.$$

Writing $m(x) = x/2$, the conditional expectation given the random variable X is $m(X)$. Thus

$$\boxed{\mathbb{E}[Y \mid X] = \frac{X}{2} \quad \text{almost surely.}}$$

The expression $\mathbb{E}[Y \mid X = x]$ gives the mean for a specified value x; $\mathbb{E}[Y \mid X]$ is itself a random variable. Values assigned outside $(0,1)$ do not affect this almost-sure identity.

(c) Tower property and direct verification. Since $0 < Y < 1$ almost surely, Y is integrable and the tower property applies:

$$\mathbb{E}[Y] = \mathbb{E}\big[\mathbb{E}[Y \mid X]\big] = \mathbb{E}[X/2] = \tfrac{1}{2}\mathbb{E}[X].$$

Using the marginal density of X,

$$\mathbb{E}[X] = \int_0^1 x\,2x\,dx = 2\left[\frac{x^3}{3}\right]_0^1 = \frac{2}{3}, \qquad \boxed{\mathbb{E}[Y] = \frac{1}{3}.}$$

Finally, the marginal density of Y gives the same result directly:

$$\mathbb{E}[Y] = \int_0^1 y\,2(1-y)\,dy = 2\left[\frac{y^2}{2} - \frac{y^3}{3}\right]_0^1 = 2\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{1}{3}.$$